

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools.
(BL4,BL5)

Assessment Tool Weightage: 30%

Annexure A

Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.

3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

1. Network Throughput and Packet Loss

ALGORITHM NetworkPerformance

BEGIN

totalDataReceived $\leftarrow$ 8000

time $\leftarrow$ 4

packetsSent $\leftarrow$ 200

packetsReceived $\leftarrow$ 180

IF time = 0 THEN

 PRINT "Error: Time cannot be zero"

 STOP

END IF

IF packetsSent = 0 THEN

 PRINT "Error: Total packets sent cannot be zero"

 STOP

END IF

throughput $\leftarrow$ totalDataReceived / time

packetLoss $\leftarrow$ packetsSent - packetsReceived

packetLossPercentage $\leftarrow$

 (packetLoss / packetsSent) $\times$ 100

PRINT "----- Network Performance Summary -----"

PRINT "Total Data Received:", totalDataReceived, "bits"

PRINT "Time:", time, "seconds"

PRINT "Packets Sent:", packetsSent

PRINT "Packets Received:", packetsReceived

PRINT "Throughput:", throughput, "bps"

PRINT "Packet Loss:", packetLossPercentage, "%"

IF throughput > 1500 AND packetLossPercentage < 10 THEN

 PRINT "Network Classification: Efficient"

ELSE

 PRINT "Network Classification: Inefficient"

END IF

END

Result

Throughput = 2000 bps

Packet Loss = 10%

Classification = Inefficient

2. Bandwidth Utilization Monitoring

ALGORITHM BandwidthMonitoring

BEGIN

links $\leftarrow$ ["Link1", "Link2", "Link3", "Link4"]

WHILE TRUE DO

 PRINT "----- Bandwidth Monitoring -----"

 FOR each link IN links DO

 usage $\leftarrow$ GET_BANDWIDTH_USAGE(link)

 STORE usage IN bandwidthArray

 PRINT link, "Utilization:", usage, "%"

 IF usage > 80 THEN

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    PRINT "Warning:", link, "is overloaded"
    PRINT "Recommendation: Switch traffic to backup link"
    backupLink ← FIND_BACKUP_LINK(link)
    IF backupLink EXISTS THEN
        PRINT "Switching traffic to:", backupLink
        SWITCH_TRAFFIC(link, backupLink)
    ELSE
        PRINT "No backup link available"
    END IF
ELSE
    PRINT link, "is operating normally"
END IF
END FOR
WAIT 10 seconds
END WHILE
END

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3. Best Communication Path Selection

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ALGORITHM BestPathSelection
BEGIN
    WHILE TRUE DO
        FOR each network link DO
            bandwidth ← GET_BANDWIDTH(link)
            delay ← GET_DELAY(link)
            utilization ← GET_UTILIZATION(link)
            status ← GET_STATUS(link)
            STORE bandwidth IN bandwidthArray
            STORE delay IN delayArray
            STORE utilization IN utilizationArray
            IF status = "FAILED" THEN
                linkScore ← 0
            ELSE
                linkScore ←
                    (0.5 × bandwidth)
                    + (0.3 × (1 / delay))
                    + (0.2 × (1 / utilization))
            END IF
            STORE linkScore IN scoreArray
        END FOR
        bestPath ← path with HIGHEST score
        IF bestPath.status = "FAILED" OR
            bestPath.utilization > 80 THEN
            PRINT "Best path is unavailable or overloaded"
            alternatePath ← path with
                next highest valid score
            IF alternatePath EXISTS THEN
                bestPath ← alternatePath
                PRINT "Alternate path selected:", bestPath
                PRINT "Alert: Path changed"
            ELSE
                PRINT "Alert: No alternate path available"
            END IF
        ELSE
            PRINT "Best communication path:", bestPath
            PRINT "Path Score:", bestPath.score
        END WHILE
    END
END

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